

Adrian Marin Mag

Data Scientist — Geophysics, Inverse Problems & Scientific Computing

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Profile

PhD candidate in geophysics at the University of Oxford, specialising in mathematical and computational methods for inverse problems, inference, and uncertainty quantification. My work focuses on extracting information about physical structure from incomplete and noisy geophysical data, with particular emphasis on statistical and probabilistic approaches, including Bayesian inversion, Backus–Gilbert methods, and set-theoretical analysis.

I design and implement Python packages that encode rigorous mathematical models while providing intuitive interfaces for scientific users. I have experience in scientific Python, optimisation, numerical modelling, and uncertainty-aware prediction, and I am interested in applying these methods to mineral exploration, critical-minerals discovery, and other real-world physical systems.

Core strengths

- **Geophysical inverse problems & uncertainty quantification:** Bayesian inversion and inference, Gaussian processes, Backus–Gilbert methods, SOLA inference, and regularised linear inversions.
- **Scientific computing:** Python, packaging, testing, documentation, Linux, and HPC workflows.
- **Physical systems modelling:** numerical modelling, adjoint-based methods, operator-based formulations, optimisation, wave propagation, and subsurface imaging (global tomography).
- **Mathematical foundations:** calculus, linear algebra, functional analysis, measure theory, probability theory, statistics.

Education

- 2022–Present **PhD in Deep Earth Imaging**, *University of Oxford*, Oxford
Thesis: *Imaging the multi-scale topography of the Earth's core.*
Advisors: Dr Paula Koelemeijer, Dr Christophe Zaroli, Dr Andrew Walker
- 2018–2022 **MSci in Geophysics (First-Class Honours, 82%)**, *University College London*, London
Thesis: *Towards building 3D global seismic tomography models using proximal methods*

Research & Software Experience

- 2022–Present **PhD Researcher, Inverse Problems & Scientific Software**, *University of Oxford*
- Developed uncertainty-aware methods for **inferring subsurface physical structure from incomplete and noisy geophysical data**.
 - Extended SOLA and related linear inference methods to estimate richer subsurface properties, including gradients, spectral components, and multi-parameter structure.
 - Built Python research software for linear inference and inversion, including operator-based model abstractions, API design, testing, documentation, and CI workflows.
 - Implemented adjoint-based workflows for computing **3D finite-frequency seismic sensitivity kernels** from AxiSEM3D outputs.
 - Worked across geophysics, applied mathematics, and software development to turn theoretical inference methods into usable scientific tools.

Publications and Manuscripts

- 2025 Mag A. M., Zaroli C., Koelemeijer P. (2025). Bridging the gap between SOLA and deterministic linear inferences in the context of seismic tomography. *Geophysical Journal International*, 242(1), ggaf131.
- In preparation Mag A. M., Al-Attar D., Isaac J., Koelemeijer P. A geometric framework for multi-parameter linear inference. Manuscript in preparation for *Geophysical Journal International*. Click for preview, password:KOBOLD
- In preparation Mag A. M., Zaroli C., Koelemeijer P., Scivier S. Property-based probabilistic linear inference in function space for geophysical inverse problems. Manuscript in preparation for *Geophysical Journal International*. Click for preview, password:KOBOLD

Selected Research Software and Projects

- pygeoinf (Python) Contributed to **pygeoinf**, a Python package for uncertainty-aware linear inference and inversion using a discretisation-agnostic, operator-centric framework. Developed methods for incorporating set-based prior information into linear inference problems and implemented the associated convex-optimisation routines. Built core abstractions for sets, support functions, polyhedral approximations, function spaces, operator covariances, and linear forms. [GitHub](#)
- SOLA-DLI (Python) Developed a Python library for linear inference with deterministic priors, including package structure, examples, documentation, and CI workflows. [GitHub](#)
- Finite-Frequency Kernels (Python) Developed adjoint-based workflows for computing volumetric and geometric finite-frequency seismic sensitivity kernels from AxiSEM3D outputs. Contributions are included as additional examples in the official AxiSEM3D [GitHub](#) repository. [GitHub](#)
- SensRay (Python) Developed code for computing ray-theoretical sensitivity kernels discretised in 1D and 3D, wrapping TauP for seismic travel-time calculations. [GitHub](#)
- Fast Marching (MATLAB) Implemented 2D wavefront propagation and ray tracing in heterogeneous media for travel-time modelling using a Fast Marching method. Undergraduate numerical-methods project. Undergraduate numerical-methods project; awarded the highest recorded mark for the course.
- Thermo-mechanical Convection (MATLAB) Implemented a 2D thermo-mechanical convection model with coupled thermal and viscous flow and temperature-dependent material properties. Undergraduate geodynamics project.

Conferences & Presentations

- 2026 **Poster:** Think First, Discretize Later. EGU.
- 2025 **Talk:** Beyond inversions: the flexibility of linear inference methods. PGRiP.
- 2025 **Poster:** SOLA inferences and normal modes. MODES/Deep Earth Meeting, Oxford–Cambridge.
- 2025 **Poster:** Combining SOLA and Deterministic Linear Inferences. Inge Lehmann Symposium, Copenhagen.
- 2024 **Poster:** Constraining Earth model properties through Backus–Gilbert SOLA inferences. UKSEDI, Leeds; BSM, Reading.
- 2023 **Poster:** Mapping the Topography of Earth’s Core. PGRiP, Edinburgh.

Teaching

- 2023–2026 **College Tutor, Mathematics, Exeter & Worcester Colleges, University of Oxford**
First- and third-year tutorials: calculus, statistics, vector calculus and continuum mechanics.
- 2024 **Python Course Demonstrator, University of Oxford**
Demonstrator for a two-day intermediate scientific Python workshop covering NumPy, SciPy, Matplotlib, VSCode, and Linux. Assisted participants with exercises, debugging, and scientific-computing workflows during hands-on sessions.
- 2025 **Python Course Demonstrator, University of Oxford**
Demonstrator for an intermediate scientific computing course for environmental scientists. Assisted participants with exercises on Python data analysis, basic software development practices (testing, modular code), and intermediate shell usage. Helped debug code and clarify workflows during hands-on sessions.
- 2026 **Python Course Demonstrator, University of Oxford**
Demonstrator for an advanced Python course for environmental scientists. Helped participants work through exercises on parallel computing (NumPy, Numba, Dask), high-performance computing systems, and analysis of large datasets using Xarray and Cartopy.
- 2020–2021 **Mathematics & Physics Tutor, Notebook Tutors, London**
Online tutoring for IB, A-level, and AP syllabi.
- 2020–2021 **Connected Learning Intern, UCL, London**
Created interactive MATLAB notebooks for a numerical methods module to improve online delivery of classes.
- 2021 **Postgraduate Teaching Assistant, UCL, London**
Developed a randomised online question bank for the Global Geophysics exam, including MATLAB scripts to generate unique datasets and solutions.

Technical Skills

Programming	Python, basic MATLAB, basic C++
Scientific Python	NumPy, SciPy, Matplotlib, Jupyter, pytest, Sphinx, Xarray, Cartopy.
Software engineering	Git, GitHub Actions, packaging, testing, documentation, CI workflows, Linux, AI-assisted development workflows
Inference & modelling	Inverse problems, Bayesian inference, Gaussian processes, uncertainty quantification, convex optimisation, adjoint methods
Geophysical computing	Seismic tomography, finite-frequency kernels, wave propagation, travel-time modelling, subsurface imaging

Honours & Awards

- 2023 Grant for SPIN workshop
- 2022 Travel grant for SEDI conference

References

Available on request.

- Dr Paula Koelemeijer Associate Professor, Department of Earth Sciences, University of Oxford, paula.koelemeijer@earth.ox.ac.uk
- Dr Christophe Zaroli Lecturer, ITES, University of Strasbourg, c.zaroli@unistra.fr
- Dr David Al-Attar Lecturer, Bullard Laboratories, University of Cambridge, da380@cam.ac.uk